

Rosenfeld–Gröbner regularization, base-field saturation, and the bad locus: why the hydrogen ansatz acquires a spurious component (a_0, a_1)

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A companion note to *A New Method of Solving PDEs*, documenting a subtlety that surfaced in the computational backing. The projection script `joca.sage` decomposes the hydrogen solution locus into **five** prime strata. The Rosenfeld–Gröbner–augmented variant `joca-rg.sage`, *as originally written*, returned only **four**, replacing two genuine strata by a single spurious component (a_0, a_1) . This note explains exactly why, reconciles it with the core theorem of the Rosenfeld–Gröbner algorithm, exhibits the saturation explicitly via `normal_form`, and records which script does what.

1 Setup: the ansatz and the projection

We seek solutions of the zero-energy hydrogen Schrödinger equation $-\frac{1}{2}\nabla^2\Psi - \frac{1}{r}\Psi = E\Psi$ (cleared of $1/r$) of the form $\Psi = \Psi(v)$, where $v = v_1x + v_2y + v_3z + v_4r$, $r = \sqrt{x^2 + y^2 + z^2}$, and Ψ satisfies a second-order linear ODE with affine coefficients. As a differential system this is the *ansatz*

$$\begin{aligned} &\Psi_x - \Psi'v_x, \quad \Psi_y - \Psi'v_y, \quad \Psi_z - \Psi'v_z, \quad \Psi'_x - \Psi''v_x, \quad \Psi'_y - \Psi''v_y, \quad \Psi'_z - \Psi''v_z, \\ &\boxed{(a_0 + a_1v)\Psi'' + (b_0 + b_1v)\Psi' + (c_0 + c_1v)\Psi}, \\ &v - (v_1x + v_2y + v_3z + v_4r), \quad r^2 - x^2 - y^2 - z^2, \end{aligned} \tag{1}$$

with the eleven constants $E, v_1, \dots, v_4, a_0, a_1, b_0, b_1, c_0, c_1$ as parameters. The **projection method** (`joca.sage`) reduces the PDE modulo (1) (Ritt reduction), collects the coefficients of the remainder as polynomials in the constants, and computes the minimal associated primes of the resulting ideal. The redundancy locus—the set of constants on which the PDE is implied by the ansatz—is

$$\begin{aligned} P_1 &= (a_0, v_4, v_3, v_2, v_1), \\ P_2 &= (c_1, c_0, b_1, b_0, a_1, a_0), \\ P_3 &= (v_1^2 + v_2^2 + v_3^2, a_1, a_0, v_4), \\ P_4 &= (v_4c_0 - b_0, Ec_0 - v_4c_1, Eb_0 - v_4^2c_1, b_1, a_1 - \tfrac{1}{2}b_0, a_0, v_3, v_2, v_1), \\ P_5 &= (v_4c_0 - b_0, v_1^2 + v_2^2 + v_3^2 - v_4^2, c_1, b_1, a_1 - b_0, a_0, E). \end{aligned} \tag{2}$$

These five primes are verified (§1.12 cross-check) to be exactly the cells on which the PDE reduces to 0 modulo the ansatz.

2 The RG-augmented variant and the spurious (a_0, a_1)

The general algorithm’s hypothesis (3) asks that the ansatz be a coherent, squarefree *regular differential system*. `joca-rg.sage` discharges it by running `RosenfeldGroebner` over the coefficient field $\mathbb{Q}(E, v_1, \dots, c_1)$ (the constants must be moved into the base field for RG to terminate—see `rg_basefield.py` and the cliff in `joca-rg-orderly.sage`). RG returns a single regular component. *Its equations are not the input*. In particular the autoreduction *squares the ODE’s initial*: the regularized ODE is

$$(a_0 + a_1 v) \cdot [(a_0 + a_1 v)\Psi'' + (b_0 + b_1 v)\Psi' + (c_0 + c_1 v)\Psi]. \quad (3)$$

If one then reduces the PDE against these regularized *equations* and prime-decomposes, the decomposition collapses to four primes: P_1 , (a_1, a_0) , P_4 , P_5 , with the single coarse (a_1, a_0) standing in for P_2 and P_3 .

(a_0, a_1) is spurious, not a coarsening. It is not that P_2, P_3 “merge”; the cell (a_0, a_1) strictly *over-claims*. Indeed $V(a_0, a_1) \supsetneq V(P_2) \cup V(P_3)$, and the true redundancy ideal $I = \bigcap_i P_i$ contains the generator $v_4 b_1$, which does *not* vanish at a generic point of $a_0 = a_1 = 0$ (take $v_4, b_1 \neq 0$). So the PDE is genuinely *not* redundant on generic $a_0 = a_1 = 0$: the reported (a_0, a_1) is an artifact of the reduction, not a solution stratum.

3 Diagnosis: the missing saturation

A regular differential chain C does not represent the bare ideal $[C]$; it represents the *saturated* ideal $[C]:H_C^\infty$, where H_C is the product of the initials and separants of C . This is the content of Theorem 28 of Boulier–Lazard–Ollivier–Petitot, *Computing representations for radicals of finitely generated differential ideals*: RG computes regular systems (A_i, S_i) with

$$\sqrt{[P_0]:S_0^\infty} = [A_1]:S_1^\infty \cap \dots \cap [A_n]:S_n^\infty. \quad (4)$$

The final autoreduction rewrites the equations *and* grows the inequation set: at each step $\bar{p} = p$ full-rem $\varphi p'$ and $S_{k+1} = S_k \cup \{i_{\bar{p}}, s_{\bar{p}}\}$ (invariant I4: $i_p, s_p \in S$ for every p). So the injected initial $(a_0 + a_1 v)$ in (3) is exactly *compensated* by $\bar{S}$ containing it: the saturation $: \bar{S}^\infty$ divides it back out, and the represented ideal $[\bar{A}]:\bar{S}^\infty$ is unchanged.

This pinpoints the bug. Boulier’s object is the pair $(\bar{A}, \bar{S})$ representing $[\bar{A}]:\bar{S}^\infty$. The method `equations()` returns only $\bar{A}$. Reducing the PDE by $\bar{A}$ and prime-decomposing computes membership in $[\bar{A}]$ —*dropping the* $: \bar{S}^\infty$. The spurious (a_0, a_1) is nothing but the locus $\bar{S} = 0$ (i.e. $a_0 + a_1 v = 0 \iff a_0 = a_1 = 0$) that the saturation was meant to delete. In one line:

$$\underbrace{[\bar{A}]:\bar{S}^\infty}_{\text{radical, 5 primes}} \quad \text{vs.} \quad \underbrace{[\bar{A}]}_{\text{radical} + \{\bar{S}=0\}=(a_0, a_1)}.$$

4 The saturation made explicit: normal_form

The chain’s own *saturation-aware* reduction, `normal_form`, displays the $: \bar{S}^\infty$ directly. It returns the normal form of the PDE as a **fraction** NUM/DEN, where the denominator is the chain separant

$$\text{DEN} = \bar{S} = (a_0 + a_1 v_\parallel)^2 - a_1^2 v_4^2 (x^2 + y^2 + z^2), \quad v_\parallel := v_1 x + v_2 y + v_3 z, \quad (5)$$

the reduced $(a_0 + a_1v)^2$ modulo $r^2 = x^2 + y^2 + z^2$ —equivalently the norm $(a_0 + a_1v)(a_0 + a_1\bar{v})$ over the quadratic relation, where $\bar{v} = v_{\parallel} - v_4r$ reflects the radial part. The membership condition is $\text{NUM} = 0$ *with* $\text{DEN} \neq 0$ —one does not zero the denominator, since $\bar{S}$ is declared a non-zero-divisor. Crucially,

$$\boxed{\text{DEN vanishes exactly on } a_0 = a_1 = 0,}$$

so $a_0 = a_1 = 0$ is a **pole** of the saturated reduction. There the normal form is $0/0$: undefined. **normal.form** therefore says *nothing* about the bad locus—neither redundant nor not. The locus $\{\text{NUM} = 0, \text{DEN} \neq 0\}$ is the redundancy *off* $a_0 = a_1 = 0$, namely the three *generic* strata P_1, P_4, P_5 . The two strata P_2, P_3 live inside the pole and are invisible to it.

5 Why the raw projection recovers all five

The difference between the two routes is what the ODE degenerates *to* on the bad locus $a_0 = a_1 = 0$:

- the **regularized** ODE (3) becomes $0 = 0$ —useless as a reduction rule, hence the genuine singularity of the chain there;
- the **raw** ODE $(a_0 + a_1v)\Psi'' + (b_0 + b_1v)\Psi' + (c_0 + c_1v)\Psi$ drops to the *first-order* rule $(b_0 + b_1v)\Psi' + (c_0 + c_1v)\Psi$, which is still usable: the leader slides from Ψ'' down to Ψ' , and reduction continues.

The raw ansatz thus covers the whole parameter space uniformly, and its plain Ritt reduction (`joca.sage`) detects P_2, P_3 along with the rest. This is why the corrected `joca-rg.sage` uses RG *only* as the coherence check (discharging hypothesis (3)) and then projects against the *raw* ansatz, recovering all five primes (2).

Reconciliation with Theorem 28. There is no contradiction. The ideal $[\bar{A}] : \bar{S}^\infty$ equals the radical (so RG is a no-op *on the ideal*). But the *saturated reduction* of that ideal carries $\bar{S}$ in the denominator, so its parametric projection is blind to $\bar{S} = 0$: it returns the three generic strata. The *raw* representation of the same radical has no such denominator, so its projection covers $\bar{S} = 0$ too—all five. To recover the bad-locus strata through the saturated route one must recurse into the pole, running **RosenfeldGroebner** over the base field *modulo* (a_0, a_1) and taking the union (the comprehensive-triangular-decomposition scheme of Chen–Moreno-Maza and Montes). The raw projection collapses that recursion into a single pass.

6 The scripts and their outputs

All scripts live in `~/Papers/NewMethod/` and run under SageMath 10.7 with François Boulier’s `DifferentialAlgebra 5.3` (`sage -python` for the `.py` files).

Script	Output	Role
<code>joca.sage</code>	5 primes, ≈ 5 s	The projection method: reduce the PDE modulo the raw ansatz, collect coefficients, prime-decompose. The faithful decomposition (2); the one the paper prints.
<code>joca-rg.sage</code>	5 primes, ≈ 26 s	Adds the hypothesis-(3) coherence check: <code>RosenfeldGroebner</code> over the base field returns one regular component, then the script projects against the <i>raw</i> ansatz. Agrees with <code>joca.sage</code> . (Its header documents why one must <i>not</i> reduce against the regularized chain.)
<code>joca-normalform.sage</code>	fraction + 4/3 primes	This note’s companion. Calls the chain’s <code>normal_form</code> ; prints NUM/DEN with $\text{DEN} = \bar{S}$; shows $\text{DEN} = 0$ on $a_0 = a_1 = 0$; decomposes $\{\text{NUM} = 0\}$ (4 primes, the (a_0, a_1) being the pole) and, after saturating by (a_0, a_1) , the 3 generic strata.
<code>rg_basefield.py</code>	1 component, ≈ 16 s	Minimal standalone demo (no Sage): builds the ring, moves the constants into the base field, runs <code>RosenfeldGroebner(ansatz, basefield=F)</code> , shows it terminates with one regular component.
<code>joca-rg-orderly.sage</code>	times out	Cliff artifact: RG with the constants left as <i>ring variables</i> under an orderly ranking. Demonstrates the blow-up is ranking-independent and that the base field is what makes RG terminate.

A related experiment lives in the separate repository `~/parametric-rg` (a from-scratch port of Fakouri–Rahmany–Basiri’s *parametric* Rosenfeld–Gröbner): `examples/joca-prg.sage` runs the §1.12 cross-check (all five primes are PDE-redundancy cells) and a direct parametric-RG attempt on the ansatz that does not terminate—the same radius-driven coefficient swell, there in the differential reduction rather than in a saturation. See that repository’s `README` for the localized diagnosis.

Summary

The component (a_0, a_1) was never in the radical. It is the bad locus B where an initial vanishes, re-admitted because the script reduced by $[\bar{A}]$ instead of $[\bar{A}] : \bar{S}^\infty$. `normal_form` exhibits the missing $:\bar{S}^\infty$ as a literal denominator, with B as its pole; and the genuine sub-strata of B are recoverable only by the raw projection (whose ODE degenerates to a usable first-order rule on B) or by recursing into B . The corrected `joca-rg.sage` therefore uses RG to certify coherence and projects raw, matching `joca.sage`’s five primes.

References

- [1] F. Boulier, D. Lazard, F. Ollivier, M. Petitot, Computing representations for radicals of finitely generated differential ideals, *Appl. Algebra Eng. Commun. Comput.* **20** (2009) 73–121 (Theo-

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- [2] F. Boulier, F. Lemaire, A. Sedoglavic, On the regularity property of differential polynomials modulo regular differential chains, *CASC 2011*, LNCS **6885**, 61–72 (the `normal_form` algorithm).
- [3] C. Chen *et al.*, Comprehensive triangular decomposition, *CASC 2007*, LNCS **4770**, 73–101 (the `recurse-into-the-bad-locus` scheme).